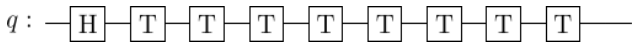
Lab 6: Single-Qubit Trajectories

1 Equator Tour

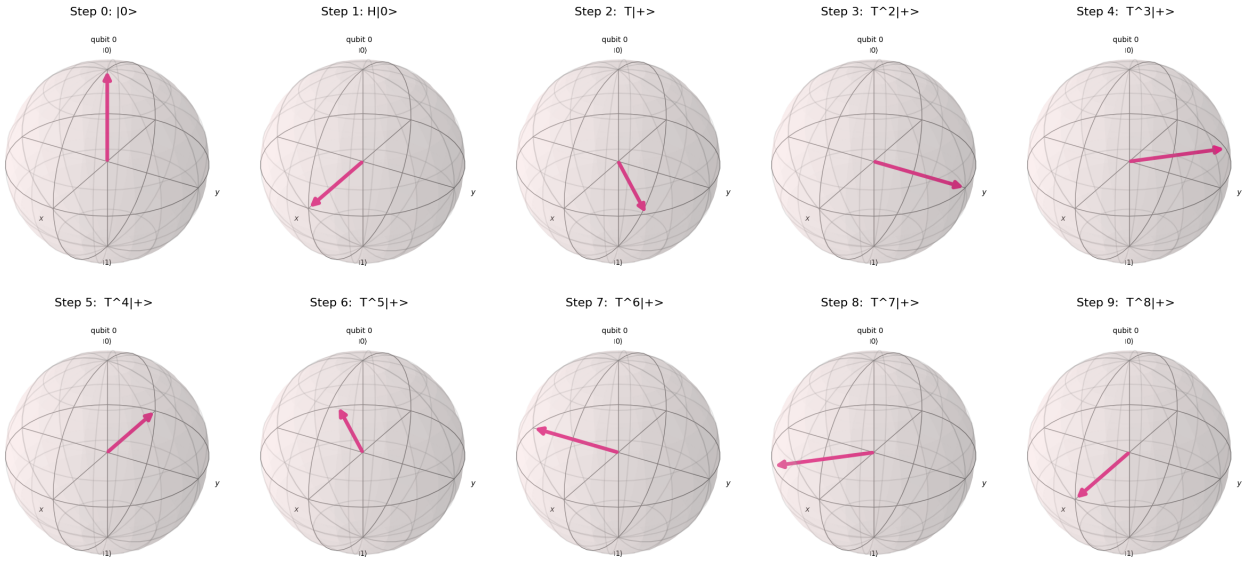
1.1 Calculations

The flowing circuit is implemented, looking at the how the state vector changes on the Bloch sphere after initialization and each gate.



Step	Bloch (x,y,z)	Angle from prev	Gate effect
$\|0\rangle$	(0,0,1)	—	Start
$\hat{H}$	(1,0,0)	90° (off z-axis)	Equator (On $+x$ axis)
$\hat{T}$	($\frac{1}{\sqrt{2}}$,$\frac{1}{\sqrt{2}}$,0)	45° (off x-axis)	Half way between $+x$ and $+y$
$\hat{T}$	(0,1,0)	45° (off x-axis)	On $+y$ axis
$\hat{T}$	(-$\frac{1}{\sqrt{2}}$,$\frac{1}{\sqrt{2}}$,0)	45° (off x-axis)	Half way between $-x$ and $+y$
$\hat{T}$	(-1,0,0)	45° (off x-axis)	On $-x$ axis
$\hat{T}$	(-$\frac{1}{\sqrt{2}}$,-$\frac{1}{\sqrt{2}}$,0)	45° (off x-axis)	Half way between $-x$ and $-y$
$\hat{T}$	(0,-1,0)	45° (off x-axis)	On $-y$ axis
$\hat{T}$	($\frac{1}{\sqrt{2}}$,-$\frac{1}{\sqrt{2}}$,0)	45° (off x-axis)	Half way between $+x$ and $-y$
$\hat{T}$	(1,0,0)	45° (off x-axis)	On $+x$ axis
Total Path	90° (off z-axis), 360° (around z-axis)		$\|+\rangle$

This makes a counter-clockwise rotation (from the view of the Bloch Sphere Plot) starting and ending on the positive x-axis.



1.2 Code

```
qc = [0,0,0,0,0,0,0,0,0,0]
state = [0,0,0,0,0,0,0,0,0,0]

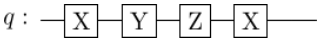
qc[0] = QuantumCircuit(1)
```

```
state[0] = Statevector(qc[0])
plot_bloch_multivector(state[0], title="Step 0: |0>")
qc[1] = qc[0].copy()
qc[1].h(0)
state[1] = Statevector(qc[1])
plot_bloch_multivector(state[1], title="Step 1: H|0>")
for i in range(2,10):
    qc[i] = qc[i-1].copy()
    qc[i].t(0)
    state[i] = Statevector(qc[i])
    if i==2:
        plot_bloch_multivector(state[i], title=f"Step {i}:  T|+>")
    else:
        plot_bloch_multivector(state[i], title=f"Step {i}:  T^{i-1}|+>")
```

2 Pauli Ladder

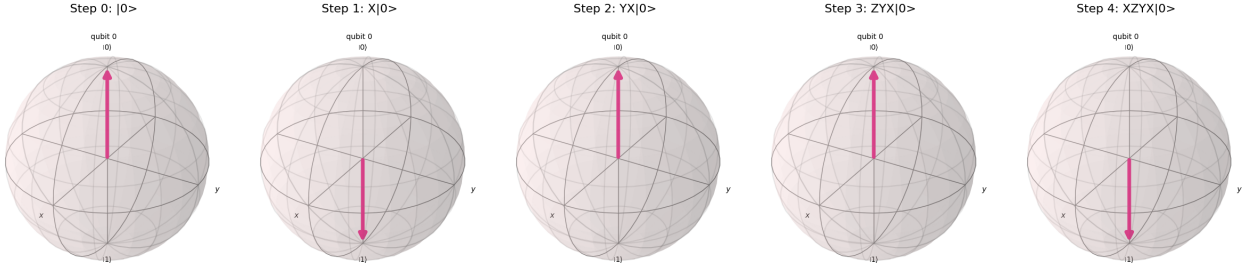
2.1 Calculations

The flowing circuit is implemented, looking at the how the state vector changes on the Bloch sphere after initialization and each gate.



Step	Bloch (x,y,z)	Angle from prev	Gate effect
$\|0\rangle$	(0,0,1)	—	Start
$\hat{X}$	(0,0,-1)	90° (off z-axis)	Equator (On $+x$ axis)
$\hat{Y}$	(0,0,1)	45° (off x-axis)	Half way between $+x$ and $+y$
$\hat{Z}$	(0,0,1)	45° (off x-axis)	On $+y$ axis
$\hat{X}$	(0,0,-1)	45° (off x-axis)	Half way between $-x$ and $+y$
Total Path	180° (off z-axis)		$\|1\rangle$

This flips the vector switching between pointing up ($|0\rangle$) to down ($|1\rangle$) in the pattern: Up, Down, Up, Up, Down.



2.2 Code

```
qc_2 = [0,0,0,0,0]
state_2 = [0,0,0,0,0]

qc_2[0] = QuantumCircuit(1)
state_2[0] = Statevector(qc_2[0])
plot_bloch_multivector(state_2[0], title="Step 0: |0>")
```

```
qc_2[1] = qc_2[0].copy()
qc_2[1].x(0)
state_2[1] = Statevector(qc_2[1])
plot_bloch_multivector(state_2[1], title="Step 1: X|0>")

qc_2[2] = qc_2[1].copy()
qc_2[2].y(0)
state_2[2] = Statevector(qc_2[2])
plot_bloch_multivector(state_2[2], title="Step 2: YX|0>")

qc_2[3] = qc_2[2].copy()
qc_2[3].z(0)
state_2[3] = Statevector(qc_2[2])
plot_bloch_multivector(state_2[2], title="Step 3: ZYX|0>")

qc_2[4] = qc_2[3].copy()
qc_2[4].x(0)
state_2[4] = Statevector(qc_2[4])
plot_bloch_multivector(state_2[4], title="Step 4: XZYX|0>")
```

3 Precision Target

3.1 Calculations

Given any vector with the form:

$$\vec{r} = r_x\hat{x} + r_y\hat{y} + r_z\hat{y}, \tag{1}$$

a vector with equal direction but with unit magnitude can be detemined with the following equation

$$\hat{r} = \frac{r_x}{|\vec{r}|}\hat{x} + \frac{r_y}{|\vec{r}|}\hat{y} + \frac{r_z}{|\vec{r}|}\hat{z}, \tag{2}$$

where

$$|\vec{r}| = \sqrt{{r_x}^2 + {r_y}^2 + {r_z}^2}. \tag{3}$$

Once the vector is normalized, it can also be represented as

$$\hat{r} = \sin(\theta)\cos(\varphi)\hat{x} + \sin(\theta)\sin(\varphi)\hat{y} + \cos(\theta)\hat{z} \tag{4}$$

From Eq. (2) and Eq. (4) equations to find both angles can be created. θ is simple,

$$\theta = \arccos\left(\frac{r_z}{|\vec{r}|}\right). \tag{5}$$

φ takes a bit more manipulation,

$$\sin(\theta)\cos(\varphi) = \frac{r_x}{|\vec{r}|}, \quad \sin(\theta)\sin(\varphi) = \frac{r_y}{|\vec{r}|},$$
$$\cos(\varphi) = \frac{r_x}{|\vec{r}|\sin(\theta)}, \quad \sin(\varphi) = \frac{r_y}{|\vec{r}|\sin(\theta)},$$

$$\tan(\varphi) = \frac{r_y}{|\vec{r}|\sin(\theta)} \times \frac{|\vec{r}|\sin(\theta)}{r_x},$$
$$\tan(\varphi) = \frac{r_y}{r_x},$$

which leads to

$$\varphi = \arctan\left(\frac{r_y}{r_x}\right). \tag{6}$$

Using the equation

$$e^{-i\frac{\varphi}{2}}\cos\left(\frac{\theta}{2}\right)|0\rangle + e^{i\frac{\varphi}{2}}\sin\left(\frac{\theta}{2}\right)|1\rangle, \tag{7}$$

a final result for the state vector can be computed to check against the rotations, as this is equivalent to

$$RZ_\phi RY_\theta |0\rangle. \tag{8}$$

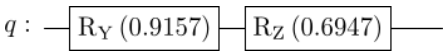
Preforming the calculations, using "np.array" with "dtype=np.longdouble" to get up to 80 bits of precision results in the following angles

$$\theta = 0.9157352247023152$$
$$\phi = 0.6947382761967033$$

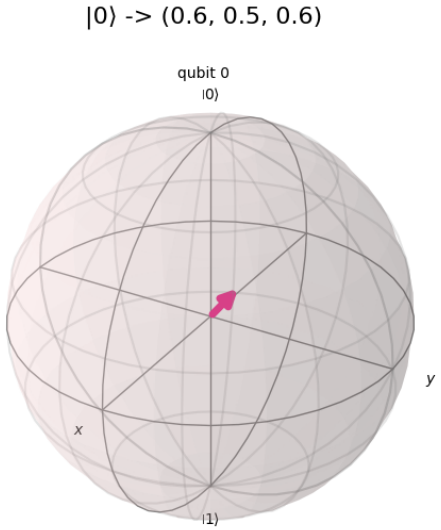
and the following state vector

$$(0.8434209057175263 - 0.3053604842456098i)|0\rangle + (0.415634232946849 + 0.15048034709754884i)|1\rangle.$$

Running the circuit below,



results in the same state vector.



3.2 Code

```
# Use dtype=np.longdouble to have up to 80-bit precision
target = np.array([.6,.5,.6], dtype=np.longdouble)
#### Normalize #####
norm = 0
for i in target:
    norm += i**2
norm = np.sqrt(norm)
normTarget = target/norm
#### Get theta ####
theta = np.arccos(normTarget[2])
#### Get phi ####
phi = np.arctan(normTarget[1]/normTarget[0])
qc_3 = QuantumCircuit(1)
qc_3.ry(theta, 0)
qc_3.rz(phi, 0)
state_3 = Statevector(qc_3)
plot_bloch_multivector(state_3, title="|0⟩ -> (0.6, 0.5, 0.6)")
```